12-March-2026 Linear Algebra — Practice Assignment

"Gauss Jordan Method"

Q:
$$3a + 2s + 4c = 106$$
$$2a + 4s + 3c = 107$$
$$4a + 3s + 2c = 102$$

Solution:-

⇒ Augmented Matrix:
$$\left[\begin{array}{ccc:c} 3 & 2 & 4 & 106 \\ 2 & 4 & 3 & 107 \\ 4 & 3 & 2 & 102 \end{array}\right]$$

• R_{13}:
$$\left[\begin{array}{ccc:c} 4 & 3 & 2 & 102 \\ 2 & 4 & 3 & 107 \\ 3 & 2 & 4 & 106 \end{array}\right] \Rightarrow \frac{1}{4}R_1: \left[\begin{array}{ccc:c} 1 & 3/4 & 1/2 & \frac{51}{2} \\ 2 & 4 & 3 & 107 \\ 3 & 2 & 4 & 106 \end{array}\right]$$

• $R_2 - 2R_1$:
• $R_3 - 3R_1$:
$$\left[\begin{array}{ccc:c} 1 & \frac{3}{4} & \frac{1}{2} & \frac{51}{2} \\ 0 & 5/2 & 2 & 56 \\ 0 & -1/4 & 5/2 & \frac{59}{2} \end{array}\right] \Rightarrow \frac{2}{5}R_2: \left[\begin{array}{ccc:c} 1 & \frac{3}{4} & 1/2 & 51/2 \\ 0 & 1 & 4/5 & \frac{112}{5} \\ 0 & -1/4 & 5/2 & 59/2 \end{array}\right]$$

$$\Rightarrow R_3 + \frac{1}{4}R_2: \left[\begin{array}{ccc:c} 1 & 3/4 & 1/2 & 51/2 \\ 0 & 1 & 4/5 & \frac{112}{5} \\ 0 & 0 & \frac{27}{10} & \frac{351}{10} \end{array}\right] \Rightarrow \frac{10}{27}R_3: \left[\begin{array}{ccc:c} 1 & 3/4 & 1/2 & \frac{51}{2} \\ 0 & 1 & 4/5 & \frac{112}{5} \\ 0 & 0 & 1 & 13 \end{array}\right]$$

$$\Rightarrow R_2 - \frac{4}{5}R_3: \left[\begin{array}{ccc:c} 1 & 3/4 & 1/2 & \frac{51}{2} \\ 0 & 1 & 0 & 12 \\ 0 & 0 & 1 & 13 \end{array}\right] \Rightarrow R_1 - \frac{1}{2}R_3: \left[\begin{array}{ccc:c} 1 & 3/4 & 0 & 19 \\ 0 & 1 & 0 & 12 \\ 0 & 0 & 1 & 13 \end{array}\right]$$

$$\Rightarrow R_1 - \frac{3}{4}R_2: \left[\begin{array}{ccc:c} 1 & 0 & 0 & 10 \\ 0 & 1 & 0 & 12 \\ 0 & 0 & 1 & 13 \end{array}\right]$$

$\boxed{a = 10}$, $\boxed{s = 12}$, $\boxed{c = 13}$